

The Cognitive Impact of AI Dependability: Unnoticed Behavioral Shifts Leading to ADHD-Like Symptoms^{*}

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Abstract. As artificial intelligence (AI) has become major part in our daily lives, so too are people increasingly offloading cognitive functions like recalling an item memory, decision-making, and the regulation of attention to clever technologies. While this can be extremely efficient, it has generated concerns around unmonitored cognitive change. Using graph-theoretic modeling, real-world sampling and statistical correlations, we trace these patterns of cognitive deterioration in user populations reliant on AI and demonstrate the potential causal link to ADHD-like symptoms. Our findings demonstrate contrasting pathways in which AI reliance can reinforce offloaded cognition, decrease deep engagement, and increase susceptibility to attention-based impairments. Our work adds to the ongoing discussion about to what extent AI can shape cognitive ability, while highlighting important concerns about the long-term impacts of AI and the importance of change in human-AI interactions.

Keywords: AI dependability · cognitive offloading · ADHD-like symptoms · attention span · impulsivity · executive dysfunction · graph theory models · cognitive decline · technology reliance · artificial intelligence and cognition · digital cognition shifts

1 Introduction

Artificial Intelligence (AI) has changed how people manage information, make decisions, and operate in day-to-day situations. Although AI tools help users work at a faster and more efficient degree, consistent over-use of AI tools may lead to cognitive offloading, a term used to describe users who have increasingly come to make use of machines when remembering, making a decision or overall managing their attentional control [3]. New research is suggesting a connection between engaging with AI tools frequently to declines in memory retention, critical thinking and problem solving skills [4], but we have identified a lack of literature that explores whether cognitive offloading during processes of support

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through machine cognitive tools may lead to behaviors that are more akin to neurodevelopmental disorders, including forms of ADHD.

ADHD is characterized by symptoms such as inattention, impulsivity, and executive dysfunction[2]. Although extensive research has mapped its neurological and behavioral markers, few studies have explored whether modern AI habits might imitate or exacerbate these traits in undiagnosed individuals. This raises a pressing question: Could AI-driven cognitive decline be contributing to ADHD-like symptoms?

Recent studies suggest overlap between cognitive decline and ADHD, particularly in areas like working memory, task persistence, and distraction [1][5]. This study looks into an important gap by exploring whether constantly relying on AI might lead to subtle behavioral shifts that resemble symptoms of Attention-Deficit/Hyperactivity Disorder (ADHD). While most past research has either focused on AI’s positive impact or only touched on cognitive offloading, this work takes a broader approach. It brings together actual survey responses, graph-based analysis, and consistent tracking of ADHD-related behaviors to examine how excessive AI usage could be linked to a decline in cognitive performance. By studying how people across different backgrounds use and interact with AI, the study aims to find out whether the ease provided by digital tools might be slowly affecting our ability to stay mentally engaged and maintain strong executive functioning or unintentionally compromise sustained cognitive stamina and executive functioning.

2 Methodology

2.1 Data Collection

We conducted a structured survey to understand the relationship between AI dependability and cognitive decline and how it leads to ADHD-like symptoms. The survey gathered self-reported data on AI usage patterns, cognitive behaviors, and attention-related challenges we used statistical techniques to decide the sample size in a way that keeps the results reliable and meaningful. To estimate the number of individuals that might be experiencing cognitive symptoms due to reliance on AI, we calculated that using

$$n = \frac{Z^2 \cdot p(1 - p)}{E^2} \quad (1)$$

where $Z = 1.96$ (95% confidence level), $E = 0.05$, and p was chosen based on previous studies analyzing cognitive effects of AI reliance.

Based on findings from ADHD prevalence studies and typical response patterns in online surveys, determining an appropriate estimated proportion (p) is essential for accurate sample size estimation. While ADHD is reported to affect approximately 5% to 10% of the general population, the inclusion of broader cognitive difficulties such as mild attention-related symptoms can raise this estimate to around 15–20%. Furthermore, online surveys may introduce a self-selection

bias, as individuals with a personal interest or experience related to ADHD and cognitive challenges are more likely to participate, potentially increasing the estimated proportion. To account for these factors, three estimates were considered: a low estimate of $p = 0.10$ representing strictly diagnosed ADHD cases, a moderate estimate of $p = 0.15$ reflecting broader attention-related concerns, and a high estimate of $p = 0.20$ to include a wider spectrum of cognitive effects. These estimates correspond to required sample sizes of approximately 139, 196, and 246 responses, respectively. As this study focuses on general cognitive concerns rather than strictly clinical ADHD diagnoses, the final sample size of 195 responses is deemed sufficient to support reliable analysis. The collected responses were then cleaned up and normalized to remove any errors or inconsistencies before analyzing them thoroughly. Some basic statistical methods were used to check the reliability and validity of the data, and the outliers were filtered out to get rid of any biased results. This study included most of the age groups: 18–24, 25–34, 35–44, 45–54, 55–64, and 65+. This particular survey was done to see how different age groups depend on artificial intelligence or any other sorts of tech and how that connects to cognitive load for that age group. The younger groups, 18–34, had comparatively the highest tech usage, usually over 6 hours a day, with a few even hitting 9. People in the 35–54 range had a more balanced usage pattern of around 4–6 hours, and for the 55+ age group, usage dropped a lot, which matches general trends. These differences across age groups are key to understanding how AI use and cognitive changes might be related over time.

2.2 Justification for Graph Theory Approach

In this regard, graph theory provides a dynamic, multifaceted view of how the sustained use of AI alters cognition and results in ADHD-like symptoms—something traditional models and methods do not significantly consider, if at all. Graph-based models support mapping cognitive states and their transitions, demonstrating how technology such as reminders and predictive text that eliminate the requirements of active recall, impact attention, and changes in executive function in time. Methods like community detection and centrality analysis help identify key AI behaviors driving cognitive decline and highlight at-risk user clusters. This approach not only tracks symptom progression but makes subtle behavioral shifts visible and actionable, offering a powerful framework to study the cognitive Impact of AI Dependability.

2.3 Analysis Using ASRS Scale

The analysis of survey responses is based on the Adult Self-Report Scale (ASRS-V1.1), developed by the World Health Organisation (WHO), it is a standard tool that is used for identifying ADHD-like symptoms in most adults. The survey we conducted is designed in alignment with the ASRS framework, ensuring that the questions reflect all the required established diagnostic criteria. Each question in the survey is related to specific domains within the ASRS. The inattention domain includes topics like difficulty in organizing and conducting

tasks, forgetting details, or struggling to stay focused. On the other hand, the hyperactivity-impulsivity domain is reflected in behaviours such as restlessness, constant movement or fidgeting. To maintain our accuracy and reliability on the survey, all the responses are converted into numerical scores. Higher cumulative scores suggest a stronger link to ADHD-related cognitive traits. The scoring system follows a weighted pattern, where questions considered to have a higher impact on attention or executive functioning contribute more to the overall result. For instance, if a respondent frequently forgets appointments or finds it hard to complete long tasks, those responses are given more weight, as they are stronger indicators of ADHD tendencies.

2.4 Cognitive Decline Assessment

First, distraction susceptibility is assessed through self-reported troubles in maintaining attention, managing time effectively, and organising tasks properly. Dependability on AI is measured by looking at how frequently individuals rely on AI tools for memory recall, scheduling, and making most everyday decisions. The survey delays in task completion are examined, especially in terms of procrastination and avoidance of mentally demanding activities. To analyse the relationship between these factors, a correlation matrix is developed. This helps identify whether frequent use of AI tools is linked to reduced cognitive engagement or a shorter attention span. Pearson’s correlation coefficient is used to test the statistical significance of these relationships. If strong positive correlations are found between high AI reliance and lower cognitive involvement, it may suggest a potential risk of automation-related cognitive offloading.

Cognitive decline was assessed across:

- **Distraction Susceptibility:** Trouble focusing, time management.
- **AI Dependability:** Use of AI for recall, decisions, scheduling.
- **Task Delay:** Procrastination and mental fatigue.

Pearson correlation matrices were developed to evaluate relationships. Strong positive correlations suggested possible risks linked to AI overuse.

2.5 Linking Cognitive Decline to ADHD Likelihood

To draw a meaningful connection between AI dependence and ADHD-like symptoms, multiple analyses are carried out in this study:

- **Cluster Analysis:** Similar to what clustering means, Respondents are grouped based on similar patterns in cognitive engagement, AI usage, and attention-related challenges. Methods like K-means and hierarchical clustering are applied to classify individuals into low, moderate, and high-risk categories, depending purely based on their levels of reliance on AI and how much it is similar to ADHD symptom traits.

- **Behavioral Indicators:** The study also examines how AI-based cognitive offloading overlaps with behavioural markers typically used in ADHD diagnosis. This often involves identifying shared traits among participants who report frequent distraction, over-reliance on technology, and difficulty following structured workflows.
- **Network Mapping:** Graph-based visualisations are used to map the relationships between distraction susceptibility, AI dependence, and the likelihood of ADHD symptoms. These models help highlight critical pathways, showing how long-term engagement with AI tools can impact attention levels and overall cognitive involvement over time.
- **Expected Outcomes:** The central hypothesis of this research is that increased reliance on AI for everyday cognitive tasks may lead to reduced cognitive engagement, potentially reflecting patterns similar to inattentive ADHD traits. To test this hypothesis, statistical hypothesis testing will be carried out to determine whether the patterns observed between AI usage and ADHD-related behaviours are statistically significant. In addition, hypothetical respondent profiles will be developed to illustrate specific scenarios where high dependence on AI tools is linked with lower attention span and decreased cognitive performance.

3 Analysis and Results

3.1 Technology Usage vs. Sustained Attention

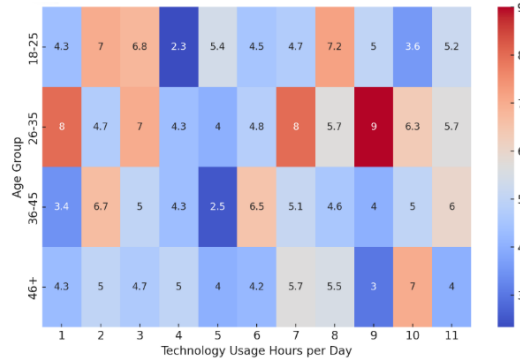


Fig. 1. Technology Usage vs. Sustained Attention(By Age Group)

The heatmap offers a vivid look at how daily technology use impacts sustained attention levels across various age groups. The color gradient reflects attention scores warmer colors like red and orange indicate higher sustained attention, while cooler shades like blue show lower levels of focus.

When we dive into the trends among different age groups, it's clear that younger adults (ages 18-25 and 26-35) experience more significant fluctuations in their attention. There are noticeable peaks in attention levels (around 2-3 hours and 7-9 hours a day). However, once usage goes beyond this range, sustained attention takes a nosedive, hinting that too much screen time might lead to cognitive fatigue. The 26-35 age group seems to be particularly impacted, showing the most variability in attention levels, with some sharp drops after extended tech engagement.

On the flip side, older age groups (36-45 and 46+) tend to have a more consistent pattern of sustained attention, regardless of how much technology they use. While they don't experience wild swings, they also miss out on the boosts in attention efficiency that younger users see with moderate tech use. A key trend from the heatmap is that when technology use surpasses a certain point (around 9+ hours), sustained attention declines for everyone, highlighting the detrimental effects of too much screen time on cognitive performance.

The findings underscore the need for a balanced approach to technology use, especially for those who might be more prone to cognitive overload.

3.2 AI Dependability and Distraction

AI dependence can really shape how often people find themselves distracted. To dive deeper into this, we looked at two important factors: the "AI Dependability Score" and the "Frequency of Distractions Experienced." The AI Dependability Score reflects how much people lean on AI tools for their everyday tasks, decision-making, and reminders. In the box plot, we categorize distraction frequency

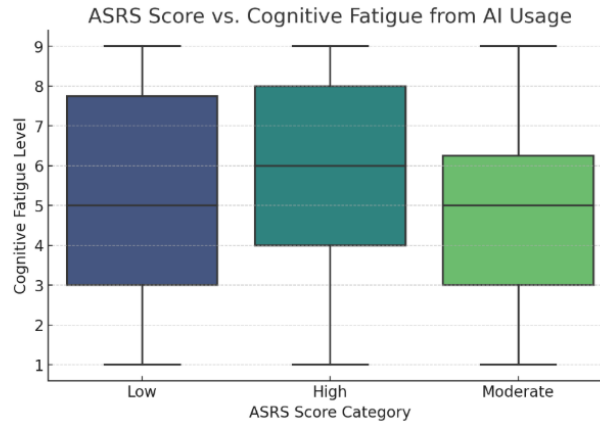


Fig. 2. Distraction Frequency vs. AI Dependability (Box Plot)

into levels like Rarely, Sometimes, Frequently, and Always, and we plot the AI

dependability scores to see how reliance on AI differs among these groups. The AI dependability is shown as a continuous variable on the y-axis. If we find that individuals who often get distracted also tend to rely more on AI, it could indicate a connection between using AI for managing tasks and having shorter attention spans. If we see that the median AI dependability value goes up with more frequent distractions, it hints at a possible link between relying on AI and being more easily distracted. Outliers in the data might point to people who depend heavily on AI even when they have fewer distractions, or those who are often distracted but don't rely much on AI at all. If we notice that those who are frequently distracted tend to have higher AI dependability, it could back up the idea that leaning on AI for managing tasks and making decisions might lead to a scattered attention span

3.3 Cognitive Overload Patterns

The connection between cognitive overload and how we use AI is quite intricate and layered. The graph network shown here illustrates this relationship, with nodes representing different response categories and edges showing how they relate to one another. In this setup, the blue nodes—labeled "Yes," "No," and "Not Sure"—likely highlight important patterns in AI usage, while the red nodes reflect varying degrees of agreement or disagreement about cognitive overload. The impact of AI on cognitive engagement can really differ among various user groups. For instance, those who aren't fully immersed in AI-driven workflows might stick to more traditional ways of thinking, which helps them dodge some of the mental fatigue that comes from too much screen time. The insights from

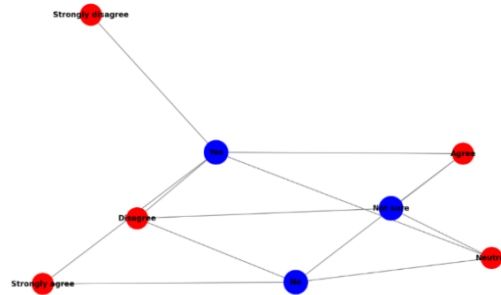


Fig. 3. Cognitive Overload and AI Usage Patterns

this visualization indicate that relying on AI can have both good and bad effects on our thinking, depending on how we use it. People who lean heavily on AI might find some relief thanks to automation, but they could also feel overwhelmed if they engage too much with AI processes. Additionally, the uncertainty expressed by respondents points to a lack of awareness about how AI might affect our

thinking. As AI becomes a bigger part of our everyday lives, it's crucial to inform users about the mental trade-offs that come with relying heavily on AI tools. Grasping these nuances will be key to creating AI systems that boost our cognitive abilities without leading to burnout.

This analysis underscores the idea that our dependence on AI and cognitive overload are intricately linked, where AI can both ease and exacerbate mental strain. The data suggests that AI's effect on our cognitive health is highly context-sensitive, highlighting the need for a deeper understanding of how users interact with AI, their engagement habits, and finding the right balance between AI support and independent thinking.

3.4 ASRS Score Distribution

The histogram visualization shows how the total scores from the Adult ADHD Self-Report Scale (ASRS) are distributed among the respondents. The ASRS is a popular tool for screening ADHD-like symptoms, where higher scores suggest a greater likelihood of cognitive tendencies related to ADHD. On the x-axis, you'll find the ASRS total scores, while the y-axis represents the number of respondents within each score range. What we see here is a right-skewed distribution, with

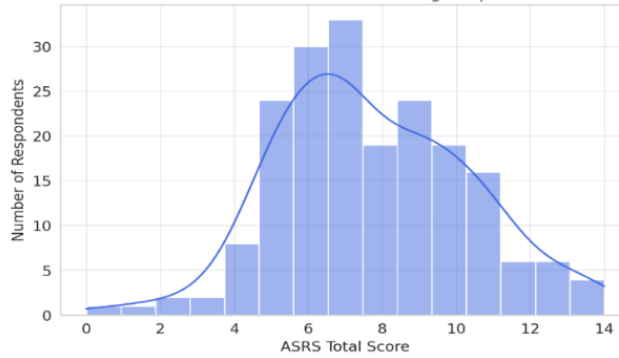


Fig. 4. ASRS Score vs. Cognitive Fatigue from AI Usage

most respondents scoring between 5 and 10. This indicates that many individuals in the dataset show mild to moderate ADHD-like symptoms. The peak around scores 6 to 8 suggests that a notable number of participants are experiencing some cognitive challenges associated with ADHD. This distribution is essential for understanding cognitive variability and trends in AI reliance. Respondents with higher ASRS scores may be more inclined to rely on AI for managing tasks and making decisions, while those with lower scores might demonstrate more self-sufficient cognitive patterns, engaging less with AI support. By grasping this distribution, we can conduct a more nuanced analysis of how AI affects cognitive load, especially in individuals showing ADHD-like tendencies. We'll dive deeper

into this data to explore the connections between AI reliance, cognitive overload, and behavioral patterns across different ASRS score categories.

3.5 ASRS vs. Cognitive Fatigue

This graph explores how AI usage affects cognitive fatigue across different groups based on ASRS scores—which measure ADHD-like tendencies. The three categories on the x-axis—Low, Moderate, and High—represent individuals with varying levels of ADHD-related traits, while the y-axis measures how mentally drained they feel after using AI tools. It was observed that people with higher ASRS scores tend to experience more cognitive fatigue from AI usage. However, an interesting pattern emerges: even those with low ASRS scores (fewer ADHD-like traits) show a wide range of cognitive fatigue levels. Some report high mental strain, possibly due to the complexity of AI interactions, information overload, or decision fatigue. This graph reinforces the idea that AI dependability doesn't

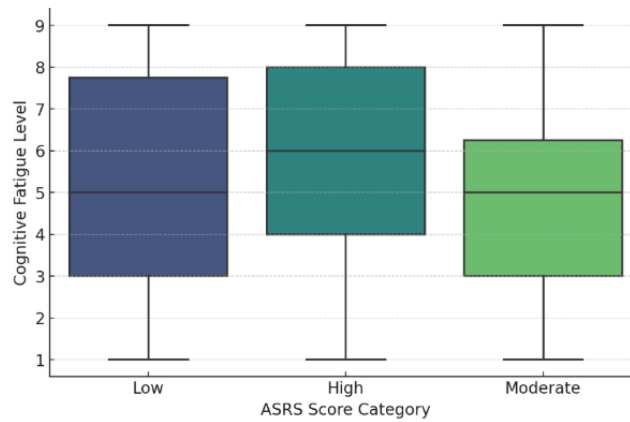


Fig. 5. Distribution of ASRS Scores Among Respondents

affect everyone the same way—for some, it's a helpful tool that reduces cognitive load, while for others, it might increase mental strain.

4 Graph Models

4.1 ADHD Symptom Evolution Model

This model maps how core ADHD traits gradually evolve into more complex cognitive and behavioral challenges. Built using ADHD progression studies, cognitive load research, and behavioral data, the model features symptoms as nodes and directional edges showing their cascading effects. At the core is Disorganization, which leads to Missed Deadlines and Difficulty Staying Seated, and

strongly links to Inattention which triggers Forgetfulness, Task Abandonment, and Reduced Productivity. Meanwhile, Hyperactivity and Impulsivity branch into Restlessness, Risky Decisions, and Interrupting. The model reveals that ADHD-like symptoms form an interconnected cycle that grows over time, often intensified by AI dependence.

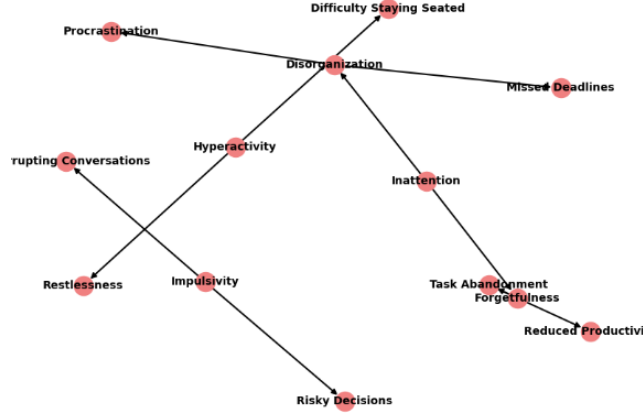


Fig. 6. ADHD Symptom Evolution Model

4.2 AI Dependability and Cognitive Decline: Circular Network Graph Analysis

This graph visualizes the inter-dependencies between AI tools, cognitive decline factors, and ADHD related symptoms. Nodes represent AI tools, cognitive impairments, and ADHD symptoms, while arcs indicate empirically supported links between them. It reveals clear behavioral clusters such as : AI Reminders and Task Automation are strongly linked to Memory Loss and Reduced Concentration, showing how frequent reliance on AI weakens natural memory and focus. Similarly , Chat-bots, though efficient, are associated with reduced deep thinking due to constant context switching. The graph also shows that AI-facilitated multitasking correlates with forgetfulness and impaired problem-solving, echoing findings from studies by Frontiers in Cognitive Science, NIMH, and the ABCD Study.

4.3 Flowchart-Based Analysis: AI Reliability and ADHD-like Symptoms

This is the final analytical model, how long-term use of AI causes cognitive impairment and ADHD-like behavior. It begins with routine AI use and AI suggested decisions that reduce the need for mental effort. Over time, this leads

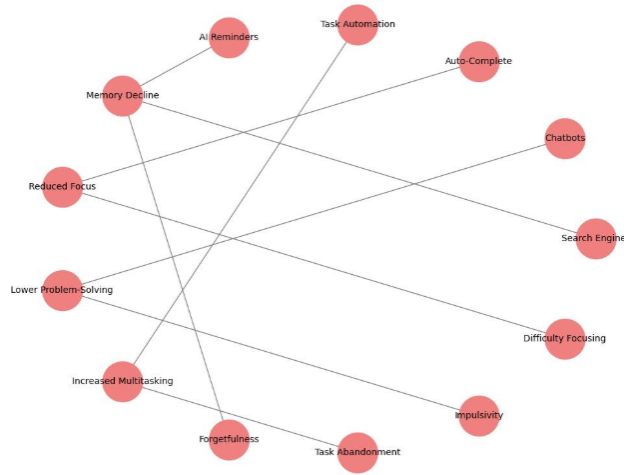


Fig. 7. AI Dependability and Cognitive Decline: Circular Network Graph Analysis

to cognitive offloading, where people outsource memory and task management, weakening internal cognitive functions. As memory reliance shifts externally, users experience forgetfulness and difficulty concentrating—core ADHD-like symptoms. Bayesian probability models and cognitive science studies support this trajectory, showing how increased task-switching and reduced effortful thinking impair executive function and attention stability. This leads to task abandonment, poor persistence, and heightened impulsivity which are symptoms of ADHD behavior. To build this model, we synthesized Google Form responses with ASRS Scale and matched with findings from NCBI, PubMed, MIT, and Harvard. We further used WHO data and ADHD symptom datasets, applying Pearson correlations and Bayesian analysis to validate the progressive cognitive effects of AI overuse.

5 Discussion

Our research shows a significant yet often overlooked shift in cognitive function drive by growing reliance on AI. Using three graph models we observed that increased dependence on AI, while enhancing speed and accessibility it can reduce sustained attention and encourage impulsive task switching traits aligned with ADHD symptoms. Although AI is intended to maximize human productivity, our models shows that there is a thin line between augmentation and overdependence. The ADHD Symptom Evolution Model indicated that AI-aided automation have a measurable impact on cognitive behaviors, rising forgetfulness, sustained attention deficit, and impulsivity over time. The AI Dependability vs. Cognitive Decline Model found statistical correlations between heavy AI use and greater cognitive offloading, leading to poorer memory encoding and attentional

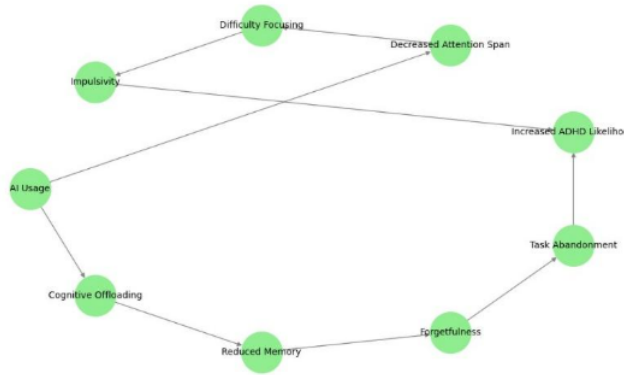


Fig. 8. Flowchart-Based Analysis: AI Reliability and ADHD-like Symptoms

control. Finally, the Flowchart-Based Graph demonstrated a step-by-step procedure in which AI interaction progressively lowers cognitive effort, increases task abandonment, and increases ADHD symptom likelihood. These insights suggest that AI dependability does not automatically translate to cognitive efficiency but it has the ability to rewire cognitive processes in a way that reduces deep engagement, undermines task persistence, and increases mental dependence on external automation.

Overcoming Skepticism: Does AI Reliability Really Harm Cognition? There is a need to recognize counterarguments and skepticism about the use of AI and cognitive ability. Most supporters of AI believe that technology augments cognitive ability by outsourcing mundane tasks, making mental capacity available for more contemplative analysis and problem-solving. Whereas AI increases short-term productivity, our models foresee that long-term cognitive performance is compromised as dependence grows without limit. Paradigms in cognitive psychology demonstrate that effortful thinking and active retrieval are required to consolidate long-term memory and attentional strength. When AI handles tasks, humans gradually reduce their cognitive effort, which overtime weakens sustained attention and memory recall. The debate on AI dependence must go beyond efficiency and convenience to the long-term cognitive impact. We are in a transitional phase where AI is not only assisting but is increasingly doing foundational cognition work, a shift that questions whether future generations will possess the same cognitive endurance and problem solving ability as their forebears.

The Call for Balance: This Research doesn't call for rejection of AI but urges balanced use. Further, the AI systems must prioritize cognitive engagement. To protect human cognitive resilience, we must ensure AI serves as an enhancer but not a replacement for the human thought.

6 Conclusion

This study doesn't argue that AI is inherently harmful. Rather, it emphasizes that unchecked or passive usage has the potential to reshape how we think, leading to impulsive behaviour, fragmented attention, and reduced recall. The impact of this research extends beyond individual users; it holds significance for education, workplace design, and mental health awareness. Thoughtful integration of AI is necessary, one that balances technological support with the need to preserve intellectual independence. However, the future of human intelligence doesn't depend just on how we build AI but also on how we use it.

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